

# User Voices, Platform Choices: Social Media Policy Puzzle with Decentralization Salt

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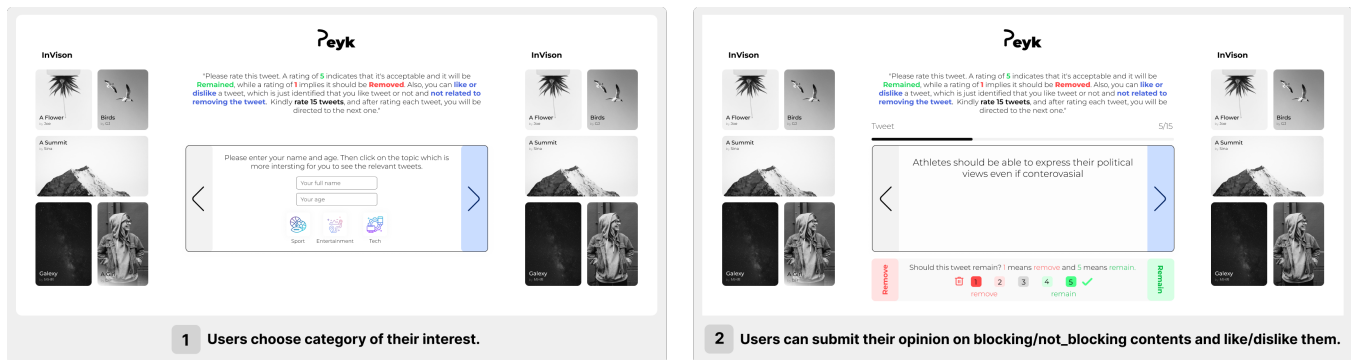


Figure 1: The implemented website for data collection about content moderation. Users can choose their category of interest among "sport," "entertainment," and "technology." They should rate from 1 to 5 for ten posts from the selected category and five posts from two other categories. Number 1 means the post should be removed, and number 5 means the post should remain.

## ABSTRACT

In the current digital era, social media platforms wield crucial influence, with the potential for biased content moderation. Considering this risk, we propose a decentralized social media policy-making in this work. The noticeable difference between people's preferences and X's established policies in a preliminary study motivates us to design a similar website to collect more comprehensive data in a diverse community. Consequently,  $N=110$  individuals from diverse backgrounds participated in our primary experiment to decide about content moderation on social media. For this purpose, 546 tweets in 3 categories are investigated, 3032 records are captured, and the effect of personal favor on content moderation is analyzed. Furthermore, we propose a novel AI-based method to learn the recommended policy of participants and achieve an accuracy of 79%. Also, by considering the suggested policy of 5 Large Language Models, it is illustrated that they cannot be the decision-makers on democratic social media platforms.

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## CCS CONCEPTS

• **Social and professional topics** → **Censorship**; User characteristics; • **Computing methodologies** → **Supervised learning by classification**; *Multi-task learning*; *Ensemble methods*; *Natural language processing*; • **Human-centered computing** → **Empirical studies in HCI**; • **Information systems** → **Social networks**; **Crowdsourcing**.

## KEYWORDS

Social Media, Social Network, Decentralization, Decentralized Policy, Content Censorship

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## 1 INTRODUCTION

These days, the widespread influence of social networks on people's lives is undeniable[38]. By increasing user engagement and audience, these platforms can shape opinions and influence people's minds based on their owners' preferences[7, 41]. The strategies employed for content moderation—ranging from purely human to fully

automated systems, with various hybrid forms in between—each come with distinct challenges and limitations[17]. Human moderators, while capable of understanding context and nuance, often struggle with the sheer volume of content, leading to potential inconsistencies and oversight[17, 38]. On the other hand, automated systems, powered by algorithms and artificial intelligence, offer scalability but can lack the subtlety required to accurately assess context, leading to erroneous content removal or retention[21, 26].

Because of the considerable power and influence, social media platforms often include various policies to be moderated. Recently, multiple policies have been explored, with most social media platforms being governed by centralized policies[42, 44].

Nowadays, an exciting trend occurs across many digital applications: a movement towards decentralization[30]. This approach offers several advantages[9], including enhanced transparency, resistance to censorship, and immutable data storage. Social media platforms, however, are also under increasing investigation as their popularity grows[40]. In venues with centralized control, narratives can be steered, content curated, and even dissent can be suppressed[33], which can lead to bias, unfair removal of content, and possible abuses[40]. In this study, we deeply explore the deviation of the users' expectation of content moderation and the policies of a prominent social media platform, referred to here as X (previously known as Twitter). We began our research with an initial survey of 89 participants, looking at their views on a selection of tweets – some blocked by X and others deemed controversial but not deleted. In preliminary findings, X's existing policies and user preferences differed by approximately 40%. This significant deviation warrants the deployment of decentralized policies to promote a more user-centric and inclusive online environment.

We designed a website based on the X aesthetic to gain deeper insight and understand the narratives. We also collected broader data and got a sense of user sentiment on whether a tweet should stay or be deleted. We recruited N=110 diverse participants for the experiment phase and examined 547 tweets related to sports, technology, and entertainment categories. In this part, participants were assigned 15 tweets based on their favor. They could attend the experiment up to 3 times, and more than 3000 records were collected. Having observed the significant deviation between user preferences and X's policies, we next aimed to assess whether the user preferences could be modeled to facilitate automatic content moderation. For this purpose, we fine-tuned a pre-trained transformer-based model on our collected data to simulate community preferences in content moderation. Our complex model achieved commendable accuracy in predicting the platform-suitability of content using advanced techniques.

In the dataset-gathering level of the experiment, we divided tweets into three categories. By categorizing topics, we can determine whether specific issues are more likely to be biased or suppressed. As a result, this approach makes it possible to highlight the challenges of moderating content across different topics in a globalized environment.

In summary, our exploration seeks to answer a pivotal question: Is it true that the dominant policies of social media reflect the sentiments of society, or are they simply the preferences of a concentrated group of people? Our findings support a decentralized

approach to social media governance and emphasize the importance of collective decision-making and increased platform fairness.

## 2 RELATED WORK

Focusing on social media studies, power centralization on social media platforms, especially, has been a subject of research and discussion for quite some time[41]. The content concentration often leads to a singular narrative and curated content, causing biases in the content presented to users. Recent studies on the implications of centralization on user perceptions revealed that centralized platforms often manifest a dominant narrative that aligns with the platform's policies and biases[19, 22]. Decentralized systems, on the other hand, have been found to result in more transparent and equitable content distribution, thereby reducing the risk of a singular narrative taking hold[16, 25, 29].

In the area of **content moderation on social media platforms**, there has been considerable concern over content moderation and its fairness[18, 39]. Research on content moderation practices has revealed the challenges and behind-the-scenes processes[3, 34]. Based on the results of our research, biases still exist in the current moderation methods despite best efforts, and the ideal of fair social media has not been achieved yet.

Transformer architectures have revolutionized natural language processing tasks. Seven years ago, the transformer model was introduced, emphasizing the self-attention mechanism that enables it to achieve state-of-the-art performance across various NLP tasks. Transformer-based models, such as BERT [12] and LaBSE [14], have since expanded possible applications of these models, including sentiment analysis, context classification, and much more. Also, a significant gap often exists between what users consider appropriate and the **platform guidelines**. Based on recent studies, involving community consensus in content moderation decisions might provide a more accurate representation of user sentiment.

It is not just social media that has been affected by the wave of decentralization. We pointed out in our introduction that **decentralization** has also been happening in areas such as finance with cryptocurrencies, education with peer assessments, and even peer-to-peer large language models (LLMs) [5, 11–13, 15, 16, 28]. Cryptocurrencies are a testament to the potential of decentralized systems[6]. The value of peer assessments in academic settings has been discussed, advocating for more decentralized methodologies in education[5, 11, 13, 32].

With the increased volume of content generated daily, manual moderation becomes impractical. As a result, some studies have been conducted targeting **automated content moderation** using machine learning[4, 36]. These years, the potential for using advanced models like GPT-3.5 [8] for content moderation has been discussed. However, as our research indicates, these state-of-the-art models sometimes align with human consensus.

**Moderating content in a globalized environment** is challenging. For instance, a benign phrase or meme in one culture could be offensive in another. As in recent studies, linguistic, cultural, and regional nuances are critical in content perception[10, 20]. This challenge highlights the importance of creating systems that can understand and adapt to diverse perspectives, ensuring fairness across the board[35].

Also, it should be mentioned that integrating **human and AI systems** can lead to complex workflows and decision-making processes that may slow down content moderation and introduce new inconsistencies[43]. Furthermore, the reliance on AI to pre-filter content for human review can inadvertently perpetuate the AI's biases or errors, affecting the overall fairness and effectiveness of the moderation process[23]. These challenges underscore the necessity for ongoing research and development in content moderation strategies, aiming to harness the advantages of both human and AI contributions while mitigating their respective limitations.

### 3 PRELIMINARY STUDY

Initially, we conducted a preliminary experiment to enhance our confidence in addressing research questions related to potential differences in individuals' viewpoints within social media in contrast to the established policies of popular social media platforms. The primary objective of this preliminary study was to measure how much public opinion differs from the policy of social media platforms. According to the core objectives, we defined a three-phased experiment. In each step, users were given a set of 15 tweets in a Google Form to assess the extent to which they believed the content should be removed from social media in a range of 0 to 10. During the first phase, we carefully selected a set of 15 tweets covering various topics characterized by diverse lengths and content. They all were chosen from content that X, formerly known as Twitter, has blocked them. Moving on to the second phase, we repeated the experiment using a new set of pre-selected tweets. These tweets were not removed from X as they aligned with X's policies. However, they were challenging due to their content. The third phase incorporated a combination of tweets, some of which had been blocked by X while others had not.

In the preliminary community selection process, we advertised the experiment to a diverse range of people with various backgrounds. This approach was essential to guarantee that the participants could represent a broad spectrum of perspectives, which led to the robustness of the results. In total, 28, 41, and 20 volunteers attended the preliminary experiment's first, second, and third phases. Individuals evaluated 15 contents in each phase, resulting in 1335 records taken together. As shown in Table 1, findings indicated a significant disparity between public opinion and the established policies of social media X (Twitter). We calculated the average rating for each tweet, followed by a center threshold of 5. Then, the results were categorized into a binary decision: whether to remove the content from social media. Our results illustrate a 46.67% alignment between public viewpoint and X's policies in the first phase. However, in the second and third phases, this metric was approximately 80.00% and 60.00%, respectively. As expected, the third phase produced results that fell between the first and second phases, reflecting its combined content selection.

We conducted an analysis to determine the degree of similarity in established policies between three distinct entities: (1) Individuals and Twitter (X), (2) Individuals and LLMS, and (3) Twitter (X) and LLMS. This assessment involved comparing the outcomes produced by each agent based on the provided content, aiming to gauge their alignment in policy implementation.

Overall, the results emphasize a significant divergence between the average preferred opinions of users and X's content censorship policies, as a similarity of less than 62.22% is demonstrated. The preliminary study results motivated our work to develop an alternative solution to propose a social media with a decentralized policy, empowering users to shape policies democratically. Therefore, it reduces the possibility of potential biases in favor of platforms' managers and owners and decreases the likelihood of censorship.

### 4 DATASET GATHERING

**Collected Tweets Dataset.** To reduce the research question to a lower scale, we decided to focus on a diverse set of three content categories, including entertainment, sports, and technology [27], which are some of the most popular categories on similar social media platforms. Additionally, we grained tweets with topics such as politics, race, and gender to increase the degree of controversy and ambiguity. All experimented tweets were created or modified by humans. We first meticulously studied the policies of similar social media platforms. Following this, we made an initial pool of tweets and categorized them as either "block" or "unblock" based on the policies of similar social media platforms. ChatGPT initially generated some unblocked tweets [8], and then we modified them to sound more ambiguous and human-like. Additionally, some other unblocked tweets and all of the blocked tweets were manually created, drawing inspiration from some famous deleted tweets and the policies of similar social media platforms. After that, we reviewed the initial pool of tweets with a closed but diverse audience and applied their feedback in order to make tweets more human-like, ambiguous, linguistically correct, and appealing to a diverse audience. Additionally, the simulated website underwent testing during this phase. Following these stages, we compiled our test data, consisting of 546 tweets, of which X will likely block 197, while 349 probably will not be blocked.

**Website Design.** We developed a website, as shown in Figure 1, to gather a diverse dataset of people's opinions on a large scale, mimicking the aesthetics of X to mitigate unintended side effects. Using Django and ReactJS in the development process. In terms of collected data, users first input their nickname and age and choose an interest category ("sport," "tech," or "entertainment"). Then, the website displays 15 tweets, consisting of 10 from the chosen category and 5 from the other two. Users rate the content from 1 (means it should be removed) to 5 (means it should remain) based on its appropriateness for social media and like or dislike it based on their opinion. We ensure liking or disliking content is distinct from voting to keep or remove it. Users can participate in the experiment up to 3 times, with backend safeguards to avoid repeating tweets in subsequent sessions.

**Experiment Process.** The experiment comprised N=110 participants aged 18 to 36 (Average = 23.97, STD = 7.27), and all had a prior basic knowledge of how X works. Considering that more than 55% of similar social media users fall between the ages of 18 and 35 [37], we decided to target this age group primarily. The participants were gathered through various means, including (1) contacting individuals with connections to the authors of the paper and inviting them to participate, (2) sending emails to researchers, and (3) utilizing diverse social media channels to target individuals of various

**Table 1: Results of the preliminary study.**

| Phase                   | Average Similarity Between People’s Preferences and X in Removing Content |
|-------------------------|---------------------------------------------------------------------------|
| First Phase             | 46.67%                                                                    |
| Second Phase            | 80.00%                                                                    |
| Third Phase             | 60.00%                                                                    |
| The Complete Experiment | 62.22%                                                                    |

professions, nationalities, genders, racial backgrounds and minorities. As mentioned, the diversity factor was heavily considered in the experiment process to have a more generalized and extendable dataset. So first, we define the fields of diversity as gender, age, nationality, race, profession, education, and sexual orientation. Then, we made a list of all possible participants and reviewed their fields of diversity, as we know. In the next step, we will group them into 20-person groups. Each group is diverse in all the mentioned fields of diversity. Then, we start asking them to participate in the experiment group by group and track the participation rate in each group to ensure a high level of diversity in the experiment. We also reminded our participants exactly one time to fill out the form in case they didn’t make any sign of filling the form.

## 5 PROPOSED METHOD

This section provides a detailed description of our proposed AI-based policy model. We take advantage of deep neural networks to design an automated policy model. One important factor is the generalization capability of these models, which means how well they can distinguish new unseen tweets. After data collection and preprocessing of raw samples, we provide embeddings of textual data by pre-trained language models that are further used as input for our estimators. Figure 2 provides an overview of the key components comprising our proposed method, which we clarify in the following paragraphs.

Raw data samples can not be directly fed into the neural networks, and a preprocessing stage is essential for preparing the raw data for analysis. The **preprocessing stage** involves removing punctuation and special characters, tokenizing words, and eliminating stop words from sentences. Additionally, we group different forms of the same word in a single one in the lemmatization step. For target labels, we first calculate the average score of each tweet across users, then make them binary by applying a mid-point threshold. Mapping tweets into a good embedding space is crucial for our primary tasks. Since our collected data is limited and using them directly for training increases the risk of overfitting, we utilize frozen language models. BERT [12] and LaBSE [14] are foundation models trained on billions of data that could provide a rich embedding space, and we use them for **embedding generation**.

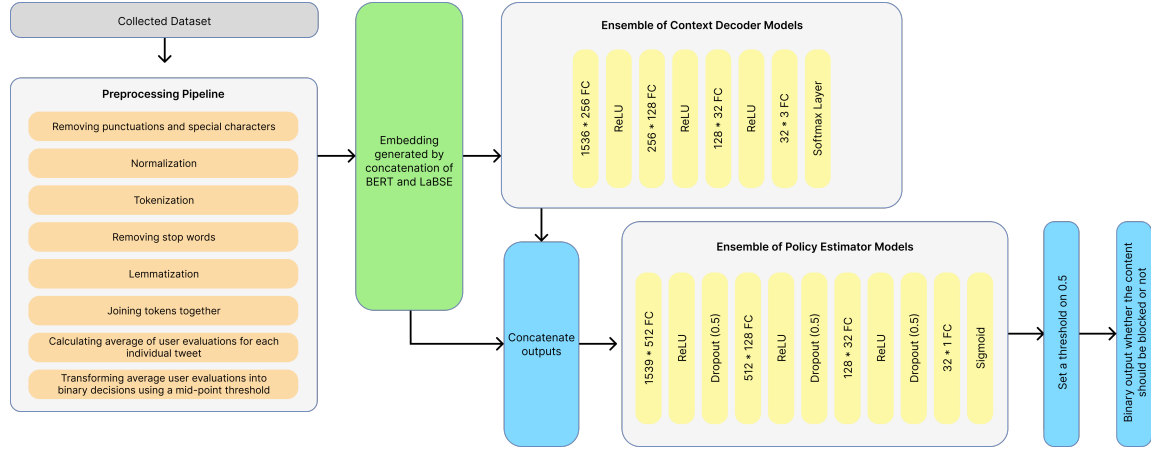
Rather than applying a single policy model to every tweet, our approach involves a two-step process. Initially, we utilize an estimator to classify tweets into predefined topics. Subsequently, we employ a specialized policy model tailored to each topic for the ultimate decision-making. We employ a Multi-Layer Perceptron (MLP) classifier as a **context decoder** and train the parameters of this network in a multi-label classification setting with cross-entropy

loss. The prediction of the context decoder is used as an extra input for the **policy estimator**. We employ specialized policy experts for different topics, forming a **mixture of policy estimators**. This means an independent estimator should be trained from scratch for each topic. Although specialized models may offer more precise results when training on extensive data samples, their performance may decline without a sufficient dataset. To alleviate this problem, we employ partially distinct estimators using the same neural network for different topics with a simple modification of its input layer. It takes both text embedding and context embedding as input.

Conditioning the policy estimator with context embedding has some advantages. First, it can exploit this extra information and make more precise context-aware decisions. Second, conditioning adds negligible context-specific parameters; most parameters remain shared between different contexts. This reduces the risk of overfitting and also increases sample efficiency by enabling knowledge transfer between topics in a multi-task learning manner. Since the topic labels are available in the training dataset, we use the one-hot vector of these labels as a context condition and concatenate them with text embeddings. We use a weighted binary cross-entropy loss to optimize the policy estimator, as the collected data is imbalanced. During inference, context conditions are estimated by the context decoder and are not necessarily in one-hot format.

Neural networks tend to make wrong predictions with high confidence [31]. So, using them for policy-making needs additional caution. In the present study, we try to circumvent this issue by capturing **epistemic uncertainty**. This kind of uncertainty reliably demonstrates the lack of knowledge about data samples. Here, we estimate this measure by employing an **ensemble of estimators** instead of a single estimator [24]. Using independent estimators in the ensemble is necessary for a reliable approximation of epistemic uncertainty. In the case of using neural networks, it is shown empirically that different initialization is enough to make estimators almost independent [24].

We leverage ensembles of estimators for both context decoder and policy model, providing additional benefits beyond mitigating overfitting. In the case of the context decoder, when a tweet contains contents about multiple topics, each ensemble member may confidently assign a high probability to a single topic. Averaging these predictions produces a soft probability vector for context conditions that assigns non-zero probabilities to relevant topics. In the case of the policy model, ensemble disagreement can be used as a proxy to prevent premature decision-making when uncertainty is considerably high. This is a step toward safe decision-making.



**Figure 2: The overall procedure for the proposed method is illustrated. In the first phase, the collected dataset is preprocessed. Subsequently, embeddings are generated using BERT and LaBSE models to capture diverse perspectives and then concatenated together. A neural network-based context decoder detects the content category in the next phase. Finally, using the embedding and context of each post, a neural network component categorizes content and applies a threshold of 0.5 to determine whether the post should be removed.**

## 6 RESULTS

**Statistical Overview of the Experiment.** As shown in Table 6 in Appendix A, the conducted experiment spanned roughly 40 days, involving  $N=110$  participants with diverse backgrounds, targeting native and non-native English speakers communities across various countries, volunteers from ages 18 to 36, and various personal backgrounds. The participants voted on 546 tweets, submitted 1911 records (average per tweet = 3.50), and liked or disliked 1121 tweets (average per tweet = 2.05). We collected 3032 reactions, including likes/dislikes (average per tweet = 5.55). Results also indicate a high level of disparity in people’s preferences and established policies of X (refer to Table 7 in Appendix A).

**Performance of the Proposed Model.** We annotated the collected dataset so as not to affect the personal opinions of the participants. The number of removed and remaining tweets differs from the annotated dataset, resulting in an imbalanced dataset. Therefore, we used weighted loss in the training procedure and provided a test set with equal samples of both groups for the final evaluation. As illustrated in Table 2, the policy estimator model achieved an accuracy of 79.82%. It is worth mentioning that in the absence of the context decoder model, the accuracy of the policy estimator was about 63.68%, approximately 15% less than the final proposed model. This result indicates the effectiveness of our multi-task learning approach. Furthermore, proposing such a model with acceptable accuracy sheds light on the possibility of proposing AI-based models to facilitate democratic policy-making on such social media platforms.

**Comparison among Users Preferences, X, and Different Large Language Models.** To examine the effectiveness of large language models in content moderation on social media, we prompted five LLMs (refer to Appendix B) in our experiment’s exact set of content. The results, as shown in Figure 3 in Appendix C, indicated that the LLMs’ suggested policies are relatively highly similar to

each other, while they all have notable differences with the participants’ preferences in content removal. However, Solar-0-70b [2] performed as the best model with a low difference among Web Search, GPT-3.5-Turbo[8], Solar-0-70b[2], GPT-3.5-Turbo-Instruct[8], and Perplexity[1]. Based on their joint distribution between people’s preferences and LLMs policies in 3, LLMs seem to have a stricter policy in filtering content by considering them inappropriate. Furthermore, as shown in Figure 4 in Appendix C and its respective table (Table 3), the suggested policy of LLMs demonstrated a slight similarity in content removal with the established policies of X.

**Investigating Potential Effects of Personal Favor in Voting Content on Social Media.** Throughout the experiment, we presented ten tweets from the category the participant had chosen and five tweets from two other categories to conduct a comprehensive meta-data analysis, comparing the average user opinions within the category of interest to those in the two not interested categories. As illustrated in Table 5, participants were more sensitive to their interested category. The result indicates that, on average, people vote for content in their interested category about 0.5 less than the two other categories. This finding highlights the dependency of people’s interests on their opinions and demonstrates that people are more sensitive about their content of interest. It necessitates sample voters from diverse communities with different backgrounds in such social media with decentralized policy. Additionally, to find the correlation between participants’ votes in content removal and their pattern of like/dislike, we analyzed their joint distribution. The results, as portrayed in Figure 5 in Appendix C, revealed that liking content can lead to higher votes, respectively, and disliking it can lead to lower votes. However, as shown in Table 4, interestingly, people can distinguish between disliking content and voting for content removal. It is evident in the existence of many individuals who strongly support the preservation of specific content on social media, even if they dislike that particular tweet.

**Table 2: Accuracy of the Policy Estimator Model.**

| Model                                        | Accuracy |
|----------------------------------------------|----------|
| Random Policy Estimator (baseline)           | 50.00%   |
| Our Policy Estimator Without Context Decoder | 63.68%   |
| Our Policy Estimator With Context Decoder    | 79.82%   |

**Table 3: Joint distribution of like/dislike and vote of a tweet based on people’s preferences.**

| LLM                                         | Web Search | GPT-3.5-Turbo | Solar-0-70b | GPT-3.5-Turbo-Instruct | Perplexity |
|---------------------------------------------|------------|---------------|-------------|------------------------|------------|
| Similarity with people’s preferred policies | 41%        | 38%           | 45%         | 38%                    | 38%        |
| Similarity with X’s policies                | 50%        | 52%           | 44%         | 54%                    | 53%        |

**Table 4: Joint distribution of like/dislike and vote of a tweet based on the collected data.**

|                           | Voted 1 | Voted 2 | Voted 3 | Voted 4 | Voted 5 |
|---------------------------|---------|---------|---------|---------|---------|
| Number of Liked Tweets    | 13      | 16      | 54      | 78      | 268     |
| Number of Disliked Tweets | 162     | 82      | 93      | 63      | 122     |

**Table 5: Differences between the average vote of users in the interested category and not-interested categories.**

| Content Type                                              | Average Votes of the Participants |
|-----------------------------------------------------------|-----------------------------------|
| Content in the Category of Interest for Every Participant | 3.00                              |
| Content in Two Other Categories for Every Participant     | 3.47                              |

## 7 DISCUSSION

**Theoretical and Practical Contributions.** In this paper, we investigated democratic social media by focusing on content removal in social media platforms. Our contributions include: (1) First, we conducted a preliminary study with 89 participants in three phases to find the potential disparity between users’ preferences and X’s policies, which motivated us to do this study; (2) Following by the initial study, we developed a website, aesthetically similar to conventional social media platforms. Employing this website, utilizing a 546 predefined set of content distributed in three categories, we did our main study with N=110 individuals to collect a comprehensive dataset, which resulted in more than 3,000 records; (3) Then, by proposing a Transformers-based model to learn the preferred policy of users, we achieved an accuracy of about 79% on validation set to predict users’ preferred policy; (4) Finally, we did various meta-data analyses on the use of large language models in content moderation on social media, the correlation between like/dislike and content removal, and effect of users’ interest on their decision making about content.

**Findings.** Our experiment results highlight the significance of democratic policy-making in social media, differing notably from traditional platform policies. This study also examines AI-based models’ effectiveness in learning democratic policy on X and the

limitations of LLMs in content moderation. We discovered that users differentiate between liking content and voting for its removal. These findings illuminate the advantages and challenges of decentralized social media policies, advancing the research community toward a democratic platform.

**Limitations and Future Work.** Our study acknowledges certain limitations while also suggesting avenues for future research. To enhance the robustness of our findings, we propose expanding the scope of our initial experiment. This could involve utilizing a larger sample of tweets, incorporating a wider range of participant age groups, and increasing the number of participants overall. Such an expansion is crucial in accurately assessing the impact of democratic policy-making on social media platforms and evaluating the generalizability of the proposed approach.

In addition, future research endeavors can focus on enhancing model accuracy through larger datasets. Furthermore, there is a pressing need to ensure fairness, particularly for marginalized groups, within the realm of democratic policy-making. Refining sampling techniques to better represent these groups could be a significant area of exploration. Moreover, the ethical dimensions of decentralized policy-making on social media platforms can be an essential direction to investigate. This includes examining potential

risks and developing strategies to mitigate harmful content while upholding democratic ideals.

Another aspect that merits attention in subsequent studies is the role of LLMs in policy formulation and content moderation. Research could delve into non-RLHF LLMs for enhancing content management. Strategies such as "Chain-of-Thought" have shown promise in improving LLM performance on specialized tasks and could be further explored in the context of policy-making [45]. Moving forward, the transition from offline data to real-world deployment with user feedback poses intriguing challenges and opportunities. Adopting interactive frameworks like contextual bandits could facilitate iterative improvements based on user interactions.

Furthermore, exploring diverse forms of interactions beyond content removal and their implications on social media platforms remains a compelling area for future inquiry. Additionally, understanding the influence of user interfaces on decision-making processes, such as voting, is essential for refining policy models. Further studies can explore the potential impact of the user interface design on voting for content removal.

Lastly, by exploring the dynamics of democratic policy-making on social media platforms, our study contributes valuable insights into the intersection of technology, user behavior, and governance structures. Understanding how users engage with democratic processes in digital spaces can determine how such platforms should be designed. As a result, this study cannot only pave the way to developing more user-centric platforms but also illuminate the obstacles and possibilities of such democratic platforms.

**Ethical Considerations and Privacy Issues.** This study prioritized ethical considerations by obtaining consent from participants at every stage. Participation in our experiments was voluntary, with no external incentives provided. To safeguard privacy, participants were required to use nicknames on the designated website, ensuring the uniqueness of the content shown in each session. Moreover, the collected dataset underwent anonymization by removing nicknames.

## 8 CONCLUSION

In this study, we investigated an innovative approach to making social media with a decentralized policy to address the limitations of social media platforms. For this purpose, after a preliminary study, we designed a platform and collected a dataset to determine the policy preferred by people compared to the X's policies and large language models (LLMs) suggested policies in content moderation. In addition, we introduced a Transformers-based model to learn the preferred policy of users, and we did various meta-analyses, including the correlation between like/dislike and voting on content, to solidify the direction for further studies. The findings of this work can be employed as guidelines and challenges for further studies in designing a democratic social media based on a decentralized policy.

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A EXPERIMENT RESULTS

Table 6: Statistical Overview of the Experiment.

| Statistic Title                          | Statistic Value                     |
|------------------------------------------|-------------------------------------|
| Total Experiment Participants            | 110                                 |
| Participants' Ages                       | 18 to 36 (Mean = 23.97, STD = 7.27) |
| Total Number of Tweets                   | 546                                 |
| Total Number of Users' Votes             | 1911                                |
| Total Number of Users' Like/Dislike      | 1121                                |
| Total Number of Reactions per Tweet      | 3032                                |
| Average Number of Votes per Tweet        | 3.50                                |
| Average Number of Like/Dislike per Tweet | 2.05                                |
| Average Number of Reactions per Tweet    | 5.55                                |

Table 7: Joint distribution of content removal by people’s preferences and X’s policies.

| Content Status                          | Number of Tweet |
|-----------------------------------------|-----------------|
| Blocked By X, Blocked by People         | 84              |
| Blocked By X, Not Blocked by People     | 45              |
| Not Blocked By X, Blocked by People     | 113             |
| Not Blocked By X, Not Blocked by People | 304             |

B PROMPTING LARGE LANGUAGE MODELS

We used a straightforward yet effective prompting strategy for the LLMs. The models were instructed to label tweets based on their appropriateness, using a binary system, '0' for inappropriate and '1' for appropriate. This Method helps us assess the model’s ability to understand nuanced social norms and language contexts. The LLMs demonstrated a notable level of comprehension and decision-making ability when presented with a few examples from the dataset, a technique known as in-context learning or few-shot learning. We also combined jail-breakers with our prompts in order to reach the deeper levels of the model’s understanding and to challenge its ability to navigate more complex or unconventional scenarios. This shows they can learn quickly, but they might not be perfect sometimes. It’s influenced by the task’s complexity and the examples’ quality. Additionally, it is observed that using simple strategies like 'Chain-of-Thought' could considerably improve the performance of LLMs on specialized tasks, and dealing with these techniques to enhance LLM-driven policies deserves a dedicated study.

C META-ANALYSES PLOTS

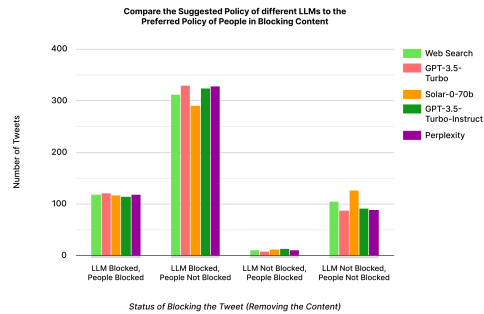


Figure 3: The comparison between people’s opinions and LLMs suggested policies.

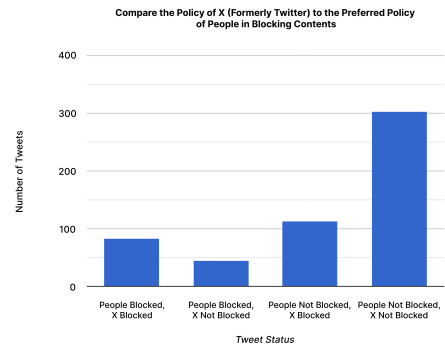


Figure 6: The comparison between people’s opinions and X’s policies.

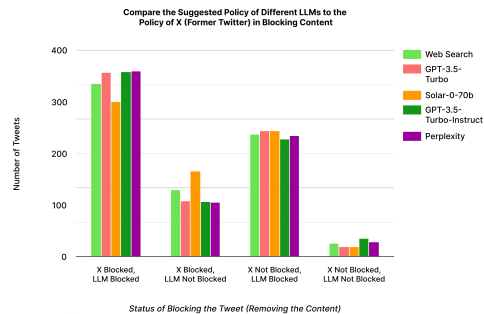


Figure 4: The comparison between X’s and LLMs suggested policies.

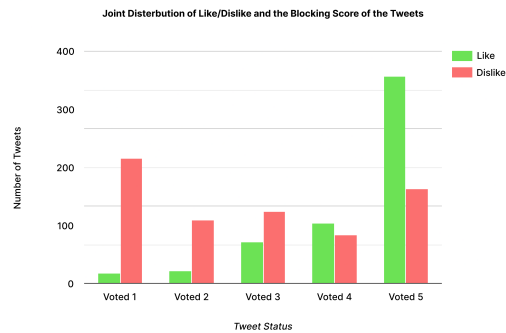


Figure 5: The correlation between like/dislike and votes of a content.